VentSense — 4S Li-ion early-warning and per-cell isolation rig

Power

File: Power.kicad_sch

Pack_Path

File: Pack_Path.kicad_sch

Measurement

File: Measurement.kicad_sch

MCU_IO

File: MCU_IO.kicad_sch

DESIGN RULES (Rev 2.1, see CLAUDE.md and notes/decisions.md)
* Fail-secure: every relay energised = connected. Relay GPIOs float at reset, so relays stay off during boot.
* One star ground: pack negative (NODE0) + logic GND + 12 V return. Heater/load returns straight to the star.
* Global nets: +12V_IN, +12V, +5V, +3V3, GND, SDA, SCL, ONEWIRE, CELL1_P-CELL4_P, RLY_SEG1-4, RLY_PACK, HTR_EN, LOAD_EN, LOAD_RTN.
* Local nets: NODE0-NODE4 (Pack path), VSENS1-4, GAS1_CO-GAS4_CO, GAS1_H2-GAS4_H2 (Measurement), SPI/buzzer/buttons (MCU).
* Modules are header symbols from VentSense_Modules.kicad_sym; verify each module pin order on arrival.
* E-stop S1 is ONE 2-unit part: unit A on Pack path (pack output), unit B on Power (heater line).

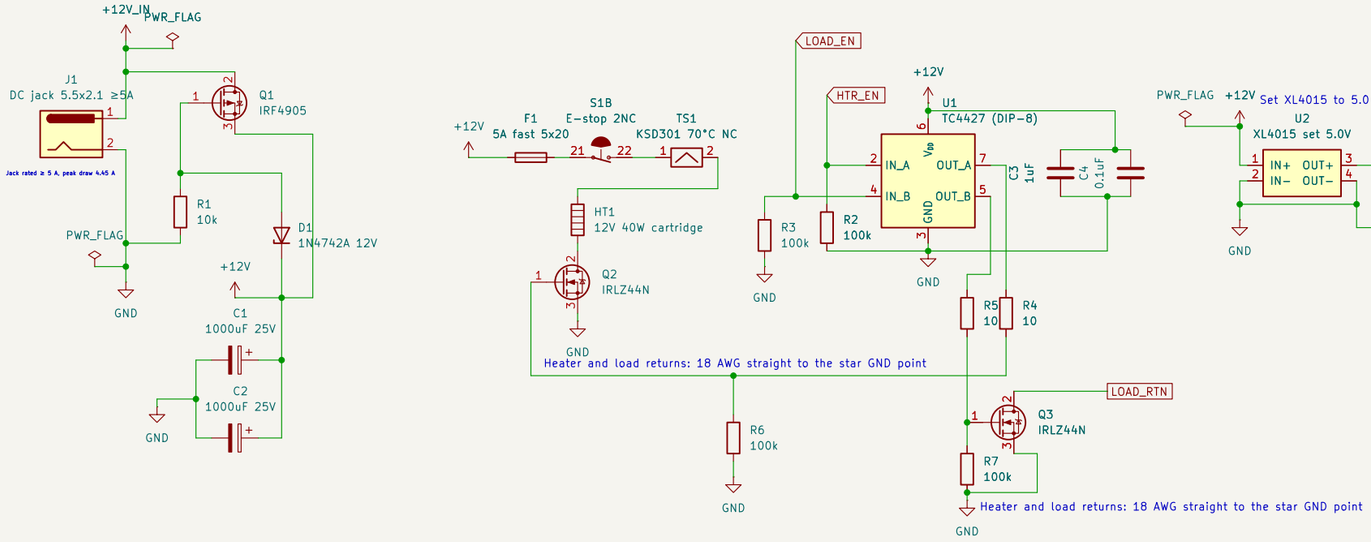
Hierarchy: Power, Pack path, Measurement, MCU and I/O
Spec: CLAUDE.md section 5 + BOM Rev 2.1 (bom/VentSense_BOM_Rev2.1.xlsx)
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Sheet: /
File: VentSense_Stage2.kicad_sch

Title: VentSense — Stage 2 schematic (root)

Size: A4	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 1/5

VentSense — Power



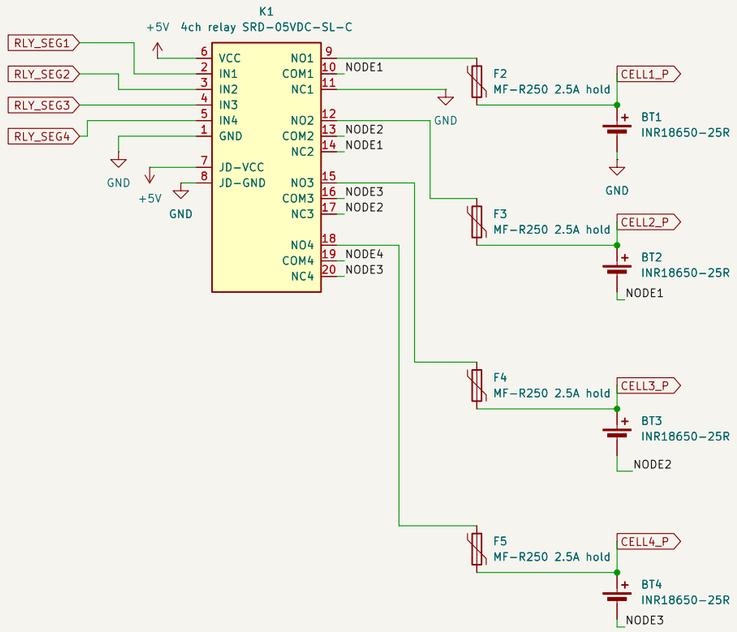
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Spec: CLAUDE.md section 5 + BOM Rev 2.1 (bom/VentSense_BOM_Rev2.1.xlsx)
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Team VentSense, IIT (BHU) Varanasi

Sheet: /Power/
File: Power.kicad_sch

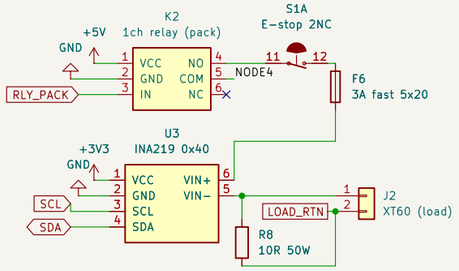
Title: VentSense — Power

Size: A3	Date: 2026-10-20	**Rev: 2.1**
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VentSense — Pack path



Pack negative (cell 1 −) is tied to logic GND at one star point.



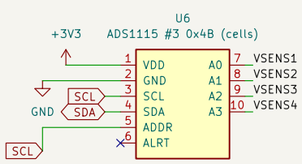
Sheet contents and values: see the notes box (Rev 2.1)
Spec: CLAUDE.md section 5 + BOM Rev 2.1 (bom/VentSense_BOM_Rev2.1.xlsx)
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Team VentSense, IIT (BHU) Varanasi

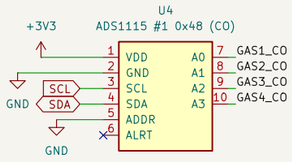
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File: Pack_Path.kicad_sch

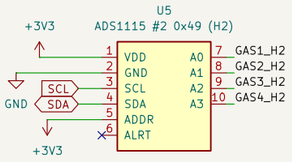
Title: VentSense — Pack path

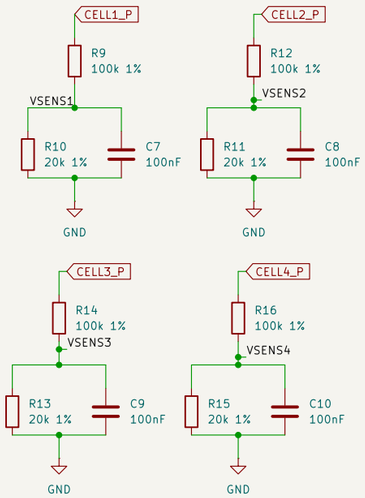
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KiCad E.D.A. 10.0.6		Id: 3/5

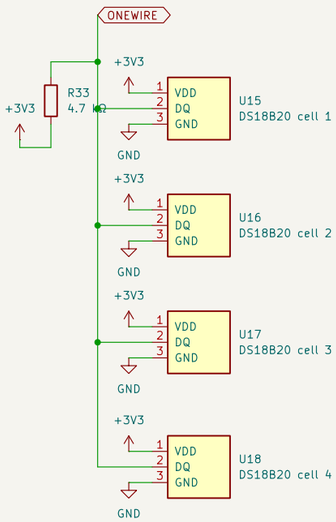
VentSense — Measurement

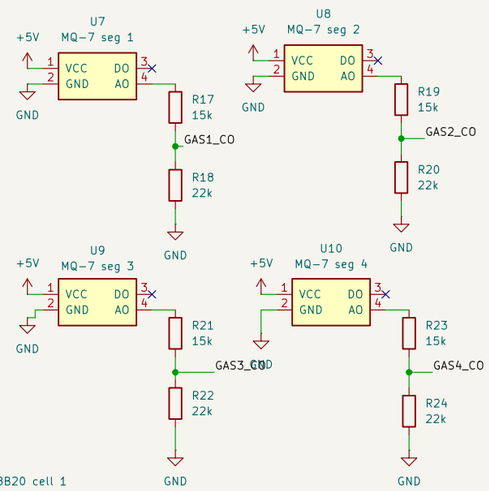


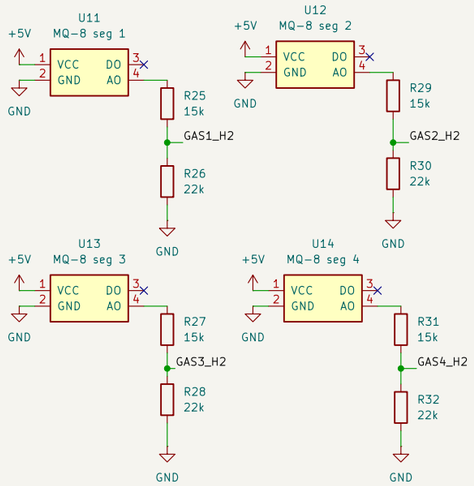












Cell divider inputs carry up to 16.8 V — keep them on their own connector on the carrier PCB.

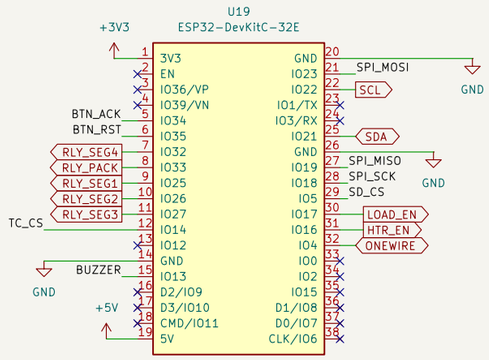
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Sheet: /Measurement/
File: Measurement.kicad_sch

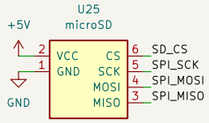
Title: VentSense — Measurement

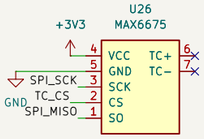
Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 4/5

VentSense — MCU and I/O

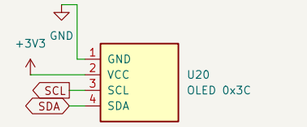


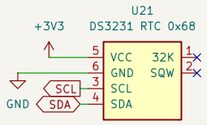


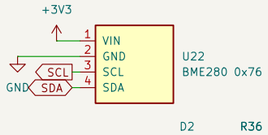


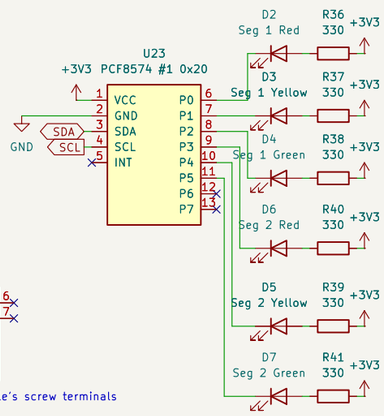


K-type thermocouple on the module's screw terminals

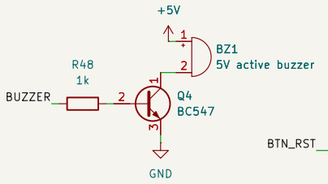


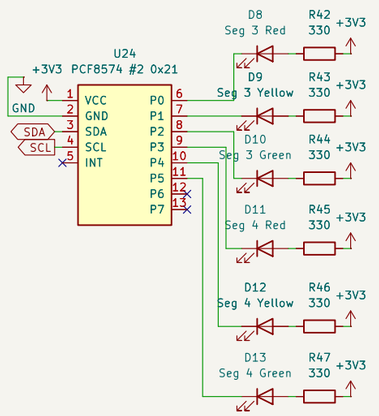






Set address jumpers on the module: #1 A2..A0 = 000 (0x20), #2 = 001 (0x21).





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VK26-IITH-SE-0279 | Vishwakarma Awards 2026-27 | Stage 2
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Sheet: /MCU_IO/
File: MCU_IO.kicad_sch

Title: VentSense — MCU and I/O

Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 5/5